

Computer Systems Security

Segurança de Sistemas de Computadores

- Master in Computer Science and Engineering
- *Mestrado em Engenharia Informática*

1st Sem. 2025/2025

Overview Presentation

- Who am I ?
- Computer Systems Security: Course Overview
 - Initial Info, resources and calendar
 - Evaluation: methods, components, rules and grading conditions
 - Bibliography: references
 - Reference on Pre-Requirements
 - Prior knowledge/background and practical experience

Who am I

Who am I

- Henrique Domingos
- Associate Prof. @ FCT/UNL
 - Computer Science Dep.
 - DI/FCT/UNL, Office P3/1
- Integrated Researcher @ NOVA LINCS
 - Computer Systems Research Group

who am I (more on academic activities)

- DI/FCT/UNL Courses - Resp. and Regency in courses such as: Computer and Network Security, Security and Privacy, Applied Cryptography, Dependable Distributed Systems, Curricular activity of activity – Engineering Project, Middleware Systems and Technologies
 - More than 80 advised MSc concluded thesis (period 2000-2025)
 - 5 Ongoing 2025/2026
 - 3 PhD students, 10 PhD Scientific committees, 18 PhD Jurys, 12 arguing member)
 - Adviser of 4 Ongoing PhD students: Scholarships (FCT/MCT, National Science Foundation) and CMU-Portugal Funding; international collaboration: Vrije Univ. CMU, CMU Cylab – IDeaS Research Centers, Amazon Research/AWS, EU
- **Academic extensions**
 - **Associated Prof. at FCT/UNL Executive School** (coordination and instructor of PostGrad courses – currently: Cybersecurity and Corporate Resilience and
 - **NOVA SBE** (regular invited courses/tutorials/seminars on E2E Encryption and Compliance, Privacy Preservation and PETs; Cloud Security and Compliance; Blockchains, FinTech
 - **Prior academic positions:** professor at Faculty of Sciences – University of Lisbon (Computer Science)
 - **Visitant professor:** Carnegie Mellon University (CMU) and Software Engineering Institute, ETH (Zurich), University of Vrije (Amsterdam), University of Rutgers

- **Examples of societal intervention:**
 - Member of the initial group (ForumIP) coordinated by Prof. Legatheaux Martins, for the introduction/interconnection of Internet in Portugal
 - See interesting references here: <https://legatheaux.eu/internet-pt/>
 - Activism/Volunteering: EFF (<https://www.eff.org/>) ISOC PT Chapter (was member of the executive board and currently activist/technical
 - Ref., <https://isoc.pt/>
- **Some industrial initiatives and activities**
 - Founder/Creation of 4 start-up companies: member of the Administration Staff, CTO and Project Manager
 - Project leadership of the first initial Internet Banking and Financing Systems in Portugal (ex., Caixa Direta OL, BESnet, BANIF, ABANCA, SIBS, Barcklays PT)
 - Consultant/Project coordinator/Auditor/Regulation and Compliance: Public Administration Sectors including (ex., GNS, CNCS, CEGER), Portuguese Government, EU – ENISA, Banking/Finance sector (including ex., Banco de Portugal, CGD, BES, SIBS, Banco do Brasil, Deutsche Bank, Barcklys, JP Morgan, Morgan&Stanley, HSBC Bank of China-PT, BNA, ValorBank, BESA, Standard Bank)

My Scientific/Research Interests

- Security and Privacy, Cybersecurity
- Dependable Computing Systems and Networks
- Byzantine-Fault/Intrusion Tolerance
- Decentralized Trust
- Privacy Preservation; Internet Censorship Avoidance/Anonymity Preservation; PETs
- Confidential computing, HW Backed Trusted Computing and Technology
- Applied Crypto
- IoT Security
- CloudSec / SecAAS
- Blockchain based Security and Privacy Preservation
- IA/LLM Security and Resilience
- OSINT-based Cyber Threat Awareness w/ AI Techniques

- Music: concluded the Higher Education Degree in Piano at the National Conservatory (Lisbon, 1987)
- Traveling / Knowledge of Different Cultures
- Sports:
 - Tennis, F1, Football
 - Handball, Rugby...
- Volunteering / Societal education on Security and Privacy

Course info, resources and calendar

- See in CLIP System: course information, syllabus and materials
 - **Characterization**: objectives, expected background/requirements, bibliography, program topics
 - **Organization**
 - Activities / classes, Evaluation rules, Grading conditions
 - **Activities (on going)**:
 - Class Materials (*Documentação de Apoio*)
 - Lectures/Lab slides, study support (ex., tests, quiz-types)
 - Proposed work assignments
 - Frequency Project-assignments
 - Evaluation results
 - **Documentation** (“on going” slides of lectures with suggested study readings, frequency tests, exams, ref. on bibliography and provided materials)
- **Very important: Awareness on CLIP notifications / CLIP Email messages**

Course activities and ECTS (corresponding effort)

Course development effort

Master Curricula in Informatics Engineering

- 6 ECTS units (28h x 6 student work ref.) = 168 h *

Includes (reference):

- Class presence (theoretical classes, lectures): $13 \times 2 = 26\text{h}$
- Lab classes: $13 \times 2 = 26\text{h}$
- Evaluation activities (tests +): $\sim 8 - 10\text{ h}$
- Other activities: student work, self-study, project assignments - development/delivery, demonstrations or complementary discussion: 106 - 108 h

* *Reference of student effort correspondence at FCT: 28h / 1 ECTS*

13 weeks (See FCT Calendar 25/26, 1st Sem.)

https://www.fct.unl.pt/sites/default/files/despacho_23_calendario_escolar_25_26.pdf

Classes and course activities: 9/Sep to 4/Dec

First week classes + 11 class weeks

Final frequency evaluations: 5/Dec to 18/Dec

Exams/Appeal: 5/Jan to 17/Jan

Setembro 2025						
S	T	Q	Q	S	S	D
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

Outubro						
S	T	Q	Q	S	S	D
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Novembro						
S	T	Q	Q	S	S	D
					1	2
3	4	5	6	7	8	9
10	11	FCT	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

Dezembro						
S	T	Q	Q	S	S	D
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

Janeiro 2026						
S	T	Q	Q	S	S	D
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Evaluation Components and Rules

Grading conditions

Mandatory components

- **2 Frequency Tests: T1 (25%), T2 (30%) : 55% of Final Eval.**
- **Final (Appeal Exam):** 55% of Final Eval.
 - for those who have Frequency Approval or want to improve final grades (substitution of frequency tests)
- **2 Project/Work Assignments: P1 (20%), P2 (25%): 45% of Final Eval)**
 - Developed individually or in group (max., 2 students/group, same lab)
 - Students/Groups must be registered up to 29/Sep
 - Registration form: <https://forms.gle/ufkrRUrTtDuia6TL6>

Optional: complementary elements (**complem. valorization**)

- Class presence and active participation (~10%)
- Solution of Proposed (Optional) Exercises (~4 as ref, ~10%.)

Tests (Individual, in PRESENCE)

- Duration ~ 2h
- Dates will be decided by the Pedagogic Commission
 - Expected*
 - T1 (middle of semester, period 25-29/Oct)
 - T2 (end of semester, period 6-10/Dec)
 - Students must register (CLIP), up to 3 days before the date of each test
- Two parts in each test
 - Part I: Closed Book Part Questions (~1h-1h15)
 - Base concepts and theoretical foundations
 - Part II: Can use “printed” or “manually written” notes, individual materials (not shared), typically ex., 6-8 printed/written pages
 - Covering problem-type, analytical and/or critical thinking questions or theoretical/practical use context
- Evaluation (0-20 scale): Max (45% Part I + 55% Part II; 55% Part I + 45% Part II)
 - rounding to one decimal place

*) Final dates and venue will be announced as soon as they are determined by the pedagogical committee and the academic services of FCT.

Final (Appeal) Exam

- Duration ~ 2h30
- Dates will be decided by the Pedagogic Commission
 - Expected in the period of exams as defined in FCT Calendar
 - **Individual, in PRESENCE**
 - In case of grade improvement: students must be registered
- Evaluation model and format: the same as that of the frequency tests.

Delivery dates will be defined and confirmed after the indication of pedagogic committee (together with the defined dates for frequency tests)

- ... Expected delivery/submission date: ~5 to 7 days before the tests
- Performed in groups (until 2 students max. of same lab group) or individually
- May include the design, implementation (involving programming), experimental evaluation criteria and reporting
- Delivery conditions:
 - Shared GitHub Repo + Summary Report
 - Submission of initial characterization of implementation aspects using a Form (Google Form)
- Evaluation
 - Coverage of requirements, correct operation, implementation quality and compliance with required delivery dates
 - Criteria (and weights) defined in each project/work assignment statement
 - For each project, students/groups can be scheduled for a practical demonstration and discussion (in presence)
 - Individualization of assessment by means of a practical/related test (performed in the same date of frequency tests)

Evaluation Rules and Grade Conditions

(See details in the CLIP “official” information)

Frequency condition (AF):

- $AF = 25\% T1 + 30\% T2 + 20\% P1 + 25\% P2$
- To obtain frequency approval condition (sufficient) :
 - $P1 \geq 7.5/20$ AND $P2 \geq 7.5/20$
 - Previously obtained frequency in SRSC in 2024/2025

Grade conditions with Frequency (NFF)

- $AF \geq 9,5/20$ AND $T1 \geq 7,5/20$ AND $T2 \geq 7,5/20$
- $NFF = 45\% AF + 25\% (T1) + 30\% (T2)$
- Grade condition with frequency:
 - $NFF \geq 9,5/20$ IF $AF \geq 9,5/20$ AND $T1 \geq 7,5/20$ AND $T2 \geq 7,5/20$

Grade conditions with final appeal exam (NF)

- Frequency approval condition (above)
- $NF = 45\% AF + 55\% \text{ Exam}$
- Grade condition with exam:
 $NF \geq 9,5/20$ AND $\text{Exam} \geq 9,5/20$

Bibliography Refs.

- W. Stallings, L. Brown, Computer Security: Principles and Practice, Pearson, 5/E (2023)
 - Ref.: <http://williamstallings.com/ComputerSecurity/>
- W. Stallings, Network Security Essentials - Applications and Services, Pearson, 6/E (2021)
 - Ref.: <http://williamstallings.com/NetworkSecurity/>
- W. Stallings, Cryptography and Network Security - Principles and Practice, Pearson 8/E (2020)
 - Ref.: <http://williamstallings.com/Cryptography/>
- P. C. van Oorschot, Computer Security and the Internet, Springer, 2/E (2021)
 - Ref.: <https://link.springer.com/book/10.1007/978-3-030-83411-1>

[WS-CS]

[WS-NSE]

**) See also information and available bibliography in CLIP*

Prog. Topics vs. Biblio.

(See Detailed Prog Topics, CLIP)

	[WS-CS]	[WS-NSE]
Overview	[WS-CS], C1	[WS-NSE], C1
1. Foundations, concepts		[WS-NSE], C2
2. Cryptography: foundations, applied crypto methods, foundations, models, techniques, algorithms, tools, constructions and correct/secure use	[WS-CS], C2 [WS-CS], C20	[WS-NSE], C3 [WS-NSE], C4
3. Authentication Access Control	[WS-CS], C3 [WS-CS], C4	
4. Net. Security, Security Protocols and Services, TCP/IP Security Stack, Security Protocols and Internet Security	[WS-CS] C22, C24, C24	[WS-NSE] C5, C6, C7, C8, C9
5. Systems Security Topics	[WS-CS], C7, C8, C9, C12	[WS-NSE], C11, C12

A. Zúquete

Segurança em Redes Informáticas, FCA, 6ª Ed (2021)

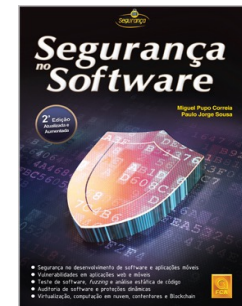
<https://www.fca.pt/pt/catalogo/informatica/seguranca-ciberseguranca-protecao-de-dados/seguranca-em-redes-informaticas-2-2/>



M. P. Correia, P. Sousa

Segurança do Software, FCA, 2ª Ed. (2017)

<https://www.fca.pt/pt/catalogo/informatica/seguranca-ciberseguranca-protecao-de-dados/seguranca-no-software-2/>



Requirements:
Prior knowledge / background
Practical requirements and experience

Previous Courses, Knowledge Base and Course Requirements (See CLIP: *Requisitos*)

Consolidation Course in the MIEI and MEI Curricula and nuclear for specialization on security and privacy specialization

Prior Knowledge Base / Background / Practical Skills

- Foundations, principles, paradigms and techniques for:
 - Computer Networks and Distributed Systems
 - Operating Systems / OS Foundations
 - Skills on use and programming on OS environments and setup/installations:
 - LINUX Distros (ex., UBUNTU, DEBIAN, ...), Linux VMs / Virtualization
 - Shell-Environment / Command-Line (console, terminals)
 - Prior skills for TCP/IP protocol stack: Internetworking programming (UDP and TCP Sockets) and Distributed Systems programming (Rest/HTTP, RST/HTTPS, Docker/Docker-Compose)
- Programming practice and programming Languages:
 - Autonomy (Programming Languages, IDEs (Eclipse, IJ, ...), **programming with command-line tools / shell environment**
 - **Languages** (may be: Java, Python, C#, C/C++ or Go)

Computer Networks, Distributed Systems

- Autonomy for Distributed Systems' Setups
- Practical skills on Programming Distributed Applications (TCP/IP Stack)

TCP/IP Appl. Programming and Java Programming/Tools

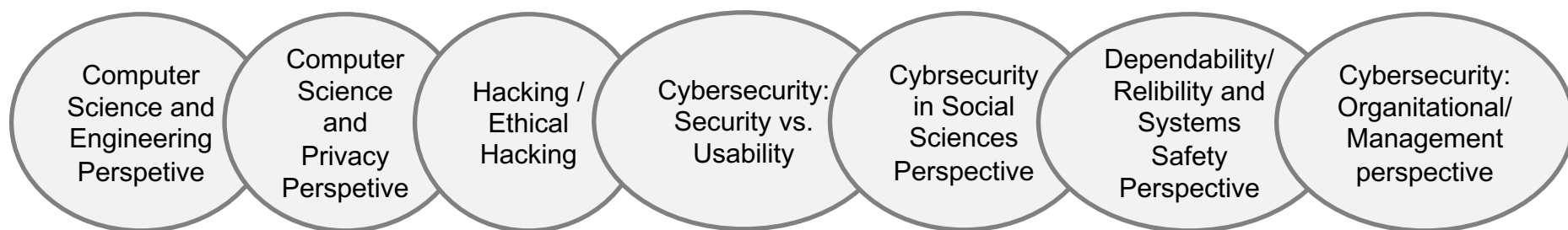
- IDEs : ex., Eclipse or IntelliJ + GitHub Repo and Code Management
- Terminal/Console: Shell Environment, Command-Line, Shell-Commands
 - MacOS or Linux (uhm ... can use VMs to have your distros)
 - Virtual Box or VMWare VMs or use of Cloud-Enabled Remote VPSs
- Java Programming and Java Tools (including console-oriented tools)
 - Refs: javac, java, keytool, javadoc, jdb, jar and jar-packaging
- Linux or MacOS Admin Basics (Installations, Setup, Monitoring Tools, ... simple shell scripting – sh, bash, shell commands)
 - Will be used/referred or demonstrated during classes
- Development w/ docker containers/images

- Network Packet Analyzers (Wireshark, tcpdump)
- Network monitoring Tools (shell)
- Virtualization: installation and setup of VM-supported environments (Linux VMs):
 - Ex, Virtual Box or VMware
 - Linux VM images, ex: Ubuntu or Debian Distros
 - Other possible distros (if you prefer)
 - Kali Distro ? Why or why not ?

Tools and installations you must consider

- Initial Installations
 - Linux (your own installation or a Cloud-enabled VM)
 - Ready with your Linux Environment (MacOS also ok)
 - Ready to use Linux (terminals, shell, command-line)
 - VirtualBox (www.virtualbox.org) or VMWare (can use as FCT student)
 - Ubuntu or Debian VMs (why/why not Kali and other “like” Distros ?)
 - Java (JDK or OpenJDK): openjdk better
 - Can use recent versions (... 17, 20, 21 ? ... or at least Java 8)
 - Java tools (Check command-line): javac, java, jar, keytool, ..
 - Important to know how to create executable JARs
 - How to build docker images (docker-compose)
 - Python (ex., 3.8 or 3.8+ is ok)
 - Eclipse (or IntelliJ or other IDE) ... *But we wil use shell-command line use (a lot)*
 - Ready for use GitHub Repo from your IDE or git shell-command
 - Setup of GitHub accounts, use w/ your IDE

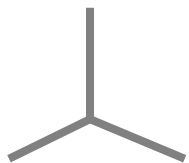
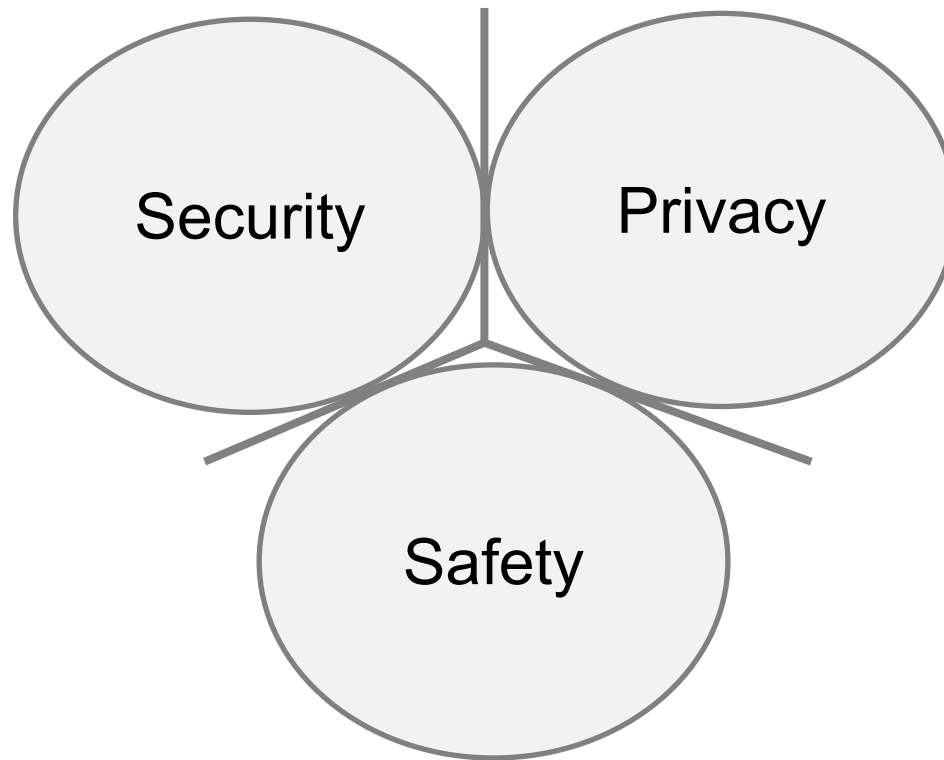
- Relevance of consolidated knowledge of security and privacy foundations, knowing the required security mechanisms, services, best security principles, practices, techniques and standards as foundational bases
- See typical coverage in suggested bibliographic references
- Today, cybersecurity covers a wide range of approaches/emphasis/perspectives/targeted concerns, examples can include:



Different cybersecurity frameworks, specializations or specialized practices, procedures, techniques and solutions, technology and tools

- CSS is a Course on Foundations (Emphasis on Security targeting security services avoiding vulnerabilities, and knowledge of how to design/implement/deploy countermeasures against potential threats and attacks considering adversarial model conditions)
 - A Computer Science and Engineering Perspective
 - It is About Foundations, Services and Mechanisms to design and operate computer systems
 - Not a course on “Hacking / Hacking Tools or Methods, Pen-Testing (even Ethical Pen-testing)”
 - Not a cybersecurity course in management or broad aspects of organizational or social perspective
 - Not targeted in “more in deep”/”specific/specialized” topics such as IoT security, Cloud-Security, AI – ML - LLMs & security,
- Security is related but different than
 - Safety
 - Privacy

Interesting possible related but different concerns



Inter-relations and Tensions

- What is security (Computer Science and Engineering) ?
- What is a secure system?
 - Do you have a “good” definition ?
- Good vs. Bad definitions
- In **computer science, security** refers to the set of measures (mechanisms, components and services) and properties that **protect systems, data, and resources from unauthorized access, modification, or disruption, with guarantees for the required security properties**
 - What security properties ?
 - What mechanisms and Services ? How to provide ?
 - We need to define this using well-accepted conceptual security frameworks

- **Safety: absence of unacceptable risks or harmful behaviors** in a system. A computer system is *safe* if it operates without causing damage to users, people, data, or the involving environment.
- A computer system is *safe* if it operates without causing damage to people, data, or the environment.
 - **Safety** → Prevents/Protects against *accidents or unintended failures* (bugs, crashes, hazards).
 - **Security** → Protects against *malicious actions* from adversarial agents (attacks as threat concretizations, unauthorized access).

Privacy is usually defined in terms of **control over information** and **protection against unauthorized access or inference**

- Privacy usually w/ many emphasis on Individual Privacy and privacy is a Human Right
 - Owned Data, Opinions, Freedom, Privacy Guarantees on using systems from third parties, Censorship Evasion, Anonymity Preservation, Protection on Abusive/Illicit Surveillance, Abusive Advertisement
- Privacy as “Secrecy Control” and “Autonomy in decision about data/information access control, Controlled confidentiality guarantees and contextual integrity conditions
- But also relevant on Privacy Engineering Systems and Tools: PETs and compliance of Privacy-Control solutions: Data-Privacy
- Deals w/ Compliance frameworks: Ethical, Deontologic, Legal
- Relevant Extensions: What about Privacy-Preserved Processing (or Computations)
 - Also related w/ Confidential Computing and Trust-Guarantees

Security vs. Privacy vs. Safety

A summary

- **Privacy:** Who can see my data/owned resources? Do I have control of this ?
- **Safety** = Will the system hurt me accidentally ? How can I have guarantees protecting me?
- **Security** = Will anyone find possible vulnerabilities to attack or misuse systems and networks I want to use with expected correct data processing or storage guarantees ? What are the countermeasures and related guarantees?"

Security vs. Privacy vs. Safety

Aspect	Definition	Focus	Key Properties / Goals	Example
Privacy	Control over one's personal or sensitive information and how it is collected, used, or shared.	Protecting information about individuals	Confidentiality, anonymity, unlinkability, data minimization	Ensuring a user's browsing history is not shared without consent
Safety	Avoidance of harmful or dangerous states in a system.	Protecting people, data, and the environment from accidents	Correctness, fault-tolerance, prevention of hazards	A program never crashes or an autonomous car avoids collisions
Security	Protection of systems and data from unauthorized access or malicious actions.	Protecting against intentional attacks	Confidentiality, integrity, availability (CIA triad)	Using encryption to prevent hackers from reading sensitive files

Questions